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Department of statistics

A Statistical Analysis of Inter Twin Prime Gaps Distribution

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Declaration

We hereby declare that the project report entitled "**A Statistical Analysis of Twin Prime Gaps Distribution**" submitted by us to the Department of Statistics, Ramakrishna Mission Residential College, is our original work.

It has been carried out under the guidance and supervision of **Dr. Sabyasachi Bhattacharya**, Professor, Department of Statistics. This work has not been submitted elsewhere for the award of any other degree or diploma.

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Abstract

Twin primes are pairs of prime numbers that differ by two, such as $(3, 5)$, $(5, 7)$, and $(11, 13)$. The study of the spacing between consecutive twin-prime pairs provides insight into the statistical structure of prime-number distributions. In this project, prime numbers were generated computationally and twin-prime pairs were extracted up to large numerical limits. Gaps between consecutive twin-prime pairs were calculated and analyzed using descriptive statistics, frequency distributions, graphical methods, probability distribution fitting, goodness-of-fit tests, and extreme value analysis.

Initially, discrete probability distributions, including the Geometric and Poisson distributions, were evaluated to assess the integer-valued nature of the gaps. Several continuous probability distributions, including the Exponential, Weibull, Gamma, and Log-normal distributions, were fitted to the observed gap data. Model performance was evaluated using likelihood-based criteria and graphical diagnostics. The Gamma distribution was found to provide the most suitable representation of the bulk behavior of the observed gaps. In addition, mean-shift analysis was performed across increasing dataset sizes to investigate scale-dependent changes in the distribution. Extreme Value Analysis (EVA) using the Peaks-Over-Threshold approach and Generalized Pareto modeling was employed to study the behavior of unusually large gaps. Regression analysis was further considered to investigate the growth pattern of twin-prime gaps with increasing prime magnitude.

The results provide a comprehensive statistical characterization of twin-prime pair gaps and contribute to understanding their distributional and asymptotic behavior.

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0.1 Introduction

Prime numbers are among the most fundamental objects in mathematics and have been studied for centuries due to their rich structure and numerous applications in number theory, cryptography, and computational mathematics. A prime number is a natural number greater than one that has exactly two positive divisors: one and itself. Despite their simple definition, the distribution of prime numbers exhibits remarkable complexity and continues to be a subject of active research.

Among the many special classes of primes, twin primes have attracted considerable attention. A pair of prime numbers $(p, p + 2)$ is called a twin-prime pair if both numbers are prime. Examples include $(3, 5)$, $(5, 7)$, $(11, 13)$, and $(17, 19)$. One of the most famous unsolved problems in mathematics, known as the Twin Prime Conjecture, asserts that there exist infinitely many twin-prime pairs. Although significant progress has been made in understanding prime gaps and the distribution of primes, a complete proof of this conjecture remains elusive.

While much of the existing literature focuses on the theoretical aspects of twin primes, statistical approaches provide an alternative perspective for investigating their behaviour. In particular, the gaps between successive twin-prime pairs offer valuable information about their distribution. As numbers become larger, twin primes appear less frequently, resulting in varying gap lengths between consecutive occurrences. Understanding the statistical properties of these gaps may provide insight into the underlying structure governing the occurrence of twin primes.

The present study examines the distribution of twin-prime gaps through computational and statistical analysis. Using R programming, prime numbers were generated and all twin-prime pairs within the dataset were identified. The gaps between successive twin-prime pairs were then calculated and analyzed using various descriptive and inferential statistical techniques. Frequency distributions, histograms, density plots, empirical cumulative distribution functions (ECDFs), box plots, and quantile–quantile (Q–Q) plots were employed to explore the characteristics of the observed data.

A major objective of this project is to investigate whether the distribution of twin-prime gaps can be adequately described by a standard probability distribution. To achieve this, graphical diagnostics and goodness-of-fit procedures were applied to assess competing distributional models. Particular attention was given to the presence of skewness, tail behaviour, and extreme observations, which are important features of gap distributions.

This work represents an interdisciplinary application of statistics to a classical problem in number theory. Rather than attempting to prove theoretical conjectures, the study focuses on extracting empirical patterns from numerical data and evaluating them through statistical methods. The results contribute to a better understanding of the observed behaviour of twin-prime gaps and provide a foundation for future research involving larger

datasets, advanced distributional modelling, and comparisons with theoretical predictions such as the Hardy–Littlewood conjectural framework.

0.2 Objectives of the Study

The primary objective of this study is to investigate the statistical behaviour of gaps between successive twin-prime pairs, examine their distributional properties, and analyse how gap characteristics evolve as prime magnitude increases.

The specific objectives are:

1. To generate twin-prime pairs and compute the corresponding gap sequence using R.
2. To examine the distributional characteristics of twin-prime gaps through descriptive and graphical analyses.
3. To investigate the variation of average gap size with increasing prime magnitude and assess potential trends using regression analysis.
4. To fit and compare suitable probability distributions for modelling the observed gap data.
5. To evaluate model performance using goodness-of-fit measures and identify the most appropriate distribution.
6. To demonstrate the application of statistical methods in the empirical study of twin-prime distributions.

The broader aim of the project is to bridge statistical analysis and number theory by developing an empirical understanding of twin-prime gap distributions and evaluating the extent to which probabilistic models can capture their observed behaviour.

0.3 Literature Review

The study of prime numbers has occupied a central position in number theory for centuries. Among the various classes of prime numbers, twin primes—pairs of primes differing by exactly two—have attracted particular attention because of their intriguing distribution and their connection to one of the most famous unsolved problems in mathematics, the Twin Prime Conjecture. This conjecture states that infinitely many twin-prime pairs exist, although a complete proof has not yet been discovered.

A significant contribution to the understanding of twin primes was made by Hardy and Littlewood through their Prime k -Tuple Conjecture. Their work provided a heuristic formula for estimating the number of twin-prime pairs less than a given number x . According to this conjecture, the density of twin primes decreases approximately as $(1/(\log x)^2)$, indicating that twin primes become increasingly rare as numbers grow larger. Despite this decreasing density, the conjecture predicts that twin primes continue to occur infinitely often.

Recent advances in prime number theory have further increased interest in prime gaps. In 2013, Yitang Zhang proved that there exist infinitely many pairs of primes separated by a bounded gap. This groundbreaking result was later refined by James Maynard and others, significantly reducing the upper bound on such gaps. Although these results do not prove the Twin Prime Conjecture, they represent major progress toward understanding the spacing between primes.

In addition to theoretical investigations, several researchers have adopted statistical approaches to study prime and twin-prime distributions. Statistical analyses often treat prime occurrences as rare events and examine the distribution of gaps between consecutive primes or twin-prime pairs. Previous studies have shown that gap distributions are highly skewed and exhibit characteristics that resemble exponential-type behaviour. Such findings suggest that probabilistic models can provide useful approximations for describing the occurrence of twin primes.

The paper “Gaps Between Consecutive Primes and the Exponential Distribution” explored the relationship between prime gaps and exponential models, demonstrating that exponential approximations can capture important features of gap distributions. Similarly, the study “Extreme Value Theory Applied to Twin Primes and Sophie Germain Prime Distributions: A Comparative Statistical Analysis” employed Extreme Value Theory (EVT) to investigate the behaviour of unusually large gaps. The authors reported that extreme gaps exhibit heavy-tail characteristics and can often be described using Generalized Extreme Value (GEV) models.

The MIT seminar paper by Daniel Berke examined twin primes from a theoretical perspective, discussing their distribution, known results, and conjectural frameworks. The paper highlighted the role of probabilistic reasoning in understanding the occurrence of twin primes and emphasized the importance of empirical observations in motivating theoretical developments.

Motivated by these studies, the present work investigates the distribution of twin-prime gaps using statistical techniques. By combining computational methods with exploratory data analysis, distribution fitting, and goodness-of-fit procedures, this project seeks to examine whether observed twin-prime gaps exhibit patterns consistent with existing theoretical and statistical models. The study therefore serves as a bridge between classical number theory and modern statistical analysis.

1. Methodology

1.1 Data Generation

Prime numbers up to a specified upper bound were generated in R using the `primes` library. This sequence formed the basis for identifying twin-prime pairs and constructing the gap sequence analyzed in this study.

1.2 Construction of Twin-Prime Gaps

Twin-prime pairs were identified as prime pairs of the form $(p, p+2)$. For two consecutive twin-prime pairs $(p_i, p_i + 2)$ and $(p_{i+1}, p_{i+1} + 2)$, the corresponding gap was defined as

$$G_i = p_{i+1} - (p_i + 2). \quad (1.1)$$

The resulting gap sequence constituted the primary dataset for subsequent analysis.

1.3 Exploratory Data Analysis

Descriptive statistics, including measures of central tendency, dispersion, skewness, and kurtosis, were computed to summarize the gap distribution. Histograms, density plots, boxplots, empirical cumulative distribution functions (ECDFs), and Q–Q plots were used to examine the empirical characteristics of the data.

1.4 Distribution Fitting and Model Assessment

The Exponential, Gamma, Weibull and Log-normal as well as discrete distribution, namely the Geometric and Poisson distributions, were fitted to the observed gaps using Maximum Likelihood Estimation (MLE). Model performance was evaluated using log-likelihood values, Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), Kolmogorov–Smirnov statistics, and graphical diagnostics.

1.5 Mean Shift Analysis

To investigate scale-dependent behaviour, twin-prime gaps were generated for progressively larger prime ranges. Separate Gamma models were fitted to each dataset, and the estimated means were compared to examine potential shifts in the distribution as the dataset size increased.

1.6 Extreme Value Analysis

The tail behaviour of the gap distribution was studied using the Peaks-Over-Threshold approach. Exceedances above selected thresholds were modelled using the Generalized Pareto Distribution (GPD), and diagnostic plots were used to assess model adequacy.

1.7 Regression Analysis

Regression models were employed to investigate the relationship between twin-prime gaps and prime magnitude. Linear and nonlinear specifications were examined to determine the most appropriate description of gap growth.

1.8 Software Environment

All computations, visualizations, and statistical analyses were performed using R (Version 4.5.0) and its associated statistical packages.

2. Dataset Description and Exploratory Data Analysis

2.1 Dataset Description

For the primary analysis, prime numbers were generated up to

$$N = 10^8. \tag{2.1}$$

All twin-prime pairs within this range were identified, and the corresponding sequence of gaps between successive twin-prime pairs was constructed. The resulting dataset contained n observed gap values, which formed the basis of the exploratory and inferential analyses presented in this study. The choice of $N = 10^8$ was motivated by the need to obtain a sufficiently large sample of twin-prime gaps while maintaining computational feasibility. The large dataset allows more reliable estimation of distributional characteristics and provides a suitable framework for studying both typical and extreme gap behaviour.

2.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to obtain an initial understanding of the statistical characteristics of the twin-prime gap sequence before formal distributional modelling.

Table 2.1: First 15 Twin Prime Pairs, Prime Values, and Their Consecutive Gaps

Index	Twin Prime Pair	Prime Value	Gap
1	(3, 5)	3	—
2	(5, 7)	5	0
3	(11, 13)	11	4
4	(17, 19)	17	4
5	(29, 31)	29	10
6	(41, 43)	41	10
7	(59, 61)	59	16
8	(71, 73)	71	10
9	(101, 103)	101	28
10	(107, 109)	107	4
11	(137, 139)	137	28
12	(149, 151)	149	10
13	(179, 181)	179	28
14	(191, 193)	191	10
15	(197, 199)	197	4
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Remark: The prime value denotes the first prime of each twin-prime pair and serves as the reference point for the corresponding gap observation. The gap between consecutive twin-prime pairs is defined as the difference between the first prime of the current twin pair and the second prime of the preceding twin pair. For example, the gap between (3, 5) and (5, 7) is $5 - 3 = 2$, while the gap between (5, 7) and (11, 13) is $11 - 7 = 4$. Only the first fifteen twin-prime pairs are displayed; the sequence continues indefinitely.

2.2.1 Descriptive Characteristics

Let

$$G_1, G_2, \dots, G_n$$

denote the observed twin-prime gaps.

The sample mean was computed as

$$\bar{G} = \frac{1}{n} \sum_{i=1}^n G_i, \quad (2.2)$$

which provides a measure of the average spacing between consecutive twin-prime pairs.

Similarly, the sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (G_i - \bar{G})^2, \quad (2.3)$$

was used to quantify the variability present in the gap sequence.

For the dataset considered in this study, the estimated mean gap was **[225.1]**, while

the estimated variance was [47018.18]. The variance substantially exceeded the mean, indicating considerable dispersion in the observed gaps.

The sample skewness was computed as

$$\text{Skewness} = \frac{1}{n} \sum_{i=1}^n \left(\frac{G_i - \bar{G}}{S} \right)^3, \quad (2.4)$$

which measures the degree of asymmetry in the distribution.

The sample kurtosis was computed as

$$\text{Kurtosis} = \frac{1}{n} \sum_{i=1}^n \left(\frac{G_i - \bar{G}}{S} \right)^4, \quad (2.5)$$

which provides information regarding the tail behaviour and peakedness of the distribution.

The estimated skewness was [1.993675], suggesting a positively skewed distribution. Furthermore, the kurtosis estimate of [9.039408] indicates the presence of a relatively long upper tail.

2.2.2 Histogram and Density Analysis

Figure 2.1 presents the histogram of the observed twin-prime gaps together with the corresponding density estimate.

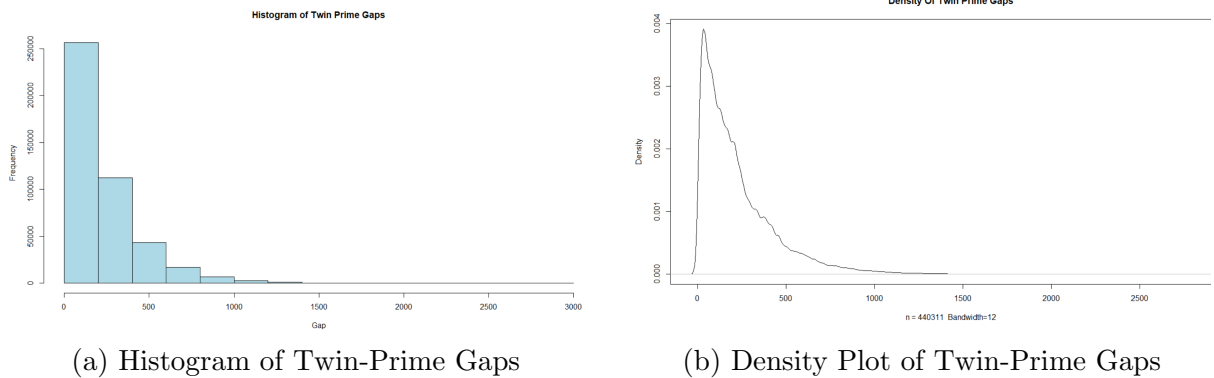


Figure 2.1: Histogram and Density Plot of Twin-Prime Gaps

The histogram of twin-prime gaps exhibits a pronounced right-skewed distribution. The majority of observations are concentrated among smaller gap values, while frequencies decrease rapidly as the gap increases. A long upper tail indicates the presence of rare but substantial separations between consecutive twin-prime pairs. Such behaviour is consistent with the decreasing density of primes and demonstrates that twin-prime gaps do not follow a symmetric distribution. And The kernel density estimate reveals a unimodal and highly right-skewed distribution. Density is concentrated among smaller gap values and declines progressively as gap size increases. The extended right tail indicates

occasional occurrence of exceptionally large twin-prime gaps. The overall shape resembles a waiting-time distribution and provides preliminary evidence supporting exponential or Weibull-type models

2.2.3 Empirical Distribution Function

The empirical cumulative distribution function (ECDF) is shown in Figure 2.2.3.

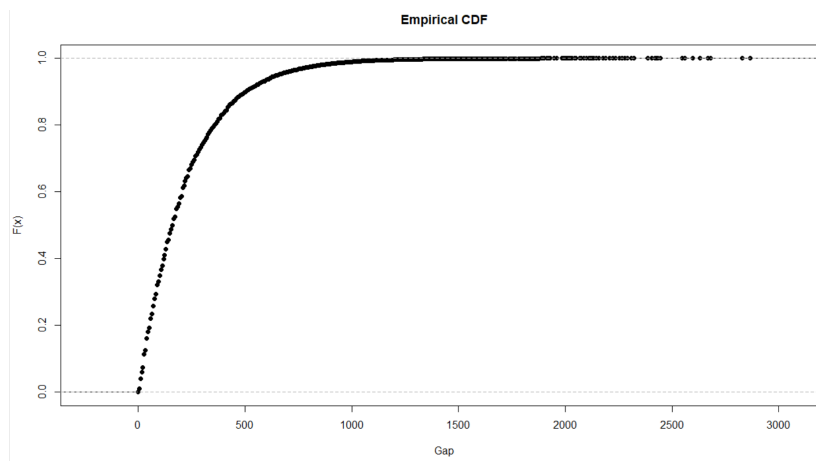


Figure 2.2: Empirical Cumulative Distribution Twin Prime Gaps

The empirical cumulative distribution function rises steeply for small gap values and subsequently approaches unity at a diminishing rate. This pattern indicates that the overwhelming majority of twin-prime gaps are relatively small, whereas large gaps occur infrequently but contribute to the extended upper tail of the distribution.

2.2.4 Boxplot Analysis

A boxplot of the twin-prime gaps is presented in Figure 2.3.

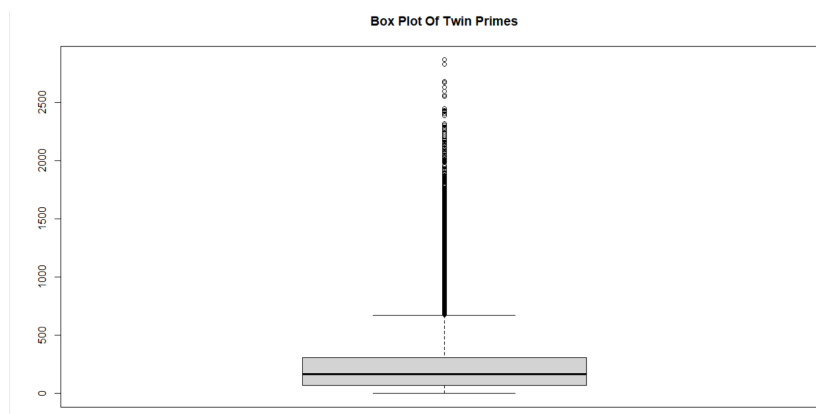


Figure 2.3: Boxplot of Twin-Prime Gaps

The box plot confirms substantial positive skewness in the twin-prime gap distribution. Numerous upper-tail outliers are observed, reflecting the presence of exceptionally large gaps between consecutive twin-prime pairs. Since these observations arise naturally from the arithmetic structure of the primes, they should be interpreted as genuine features of the distribution rather than anomalous measurements.

2.2.5 Motivation for Log-Frequency Analysis

One of the principal objectives of this study is to investigate how the frequency of twin-prime gaps changes as the gap size increases. In many naturally occurring counting processes, frequencies decline approximately according to an exponential law of the form

$$f(g) = Ce^{-\lambda g}, \quad (2.6)$$

where g denotes the gap size, $C > 0$ is a normalizing constant, and $\lambda > 0$ represents the rate of decay.

Taking the natural logarithm of both sides yields

$$\log f(g) = \log C - \lambda g. \quad (2.7)$$

Equation (5.3) implies that if the observed frequencies follow an exponential decay pattern, then a plot of $\log f(g)$ against the gap size g should exhibit an approximately linear relationship with negative slope $-\lambda$.

The log-frequency representation therefore serves as a useful graphical diagnostic for assessing whether the occurrence of twin-prime gaps decreases exponentially with increasing gap size. Deviations from linearity may indicate the presence of alternative distributional behaviour, such as heavier tails or power-law characteristics.

Accordingly, a logarithmic frequency plot was constructed to examine the empirical decay pattern of the observed twin-prime gaps.

2.2.6 Log-Frequency Analysis

To examine the decay pattern of the observed frequencies, a logarithmic frequency plot was constructed.

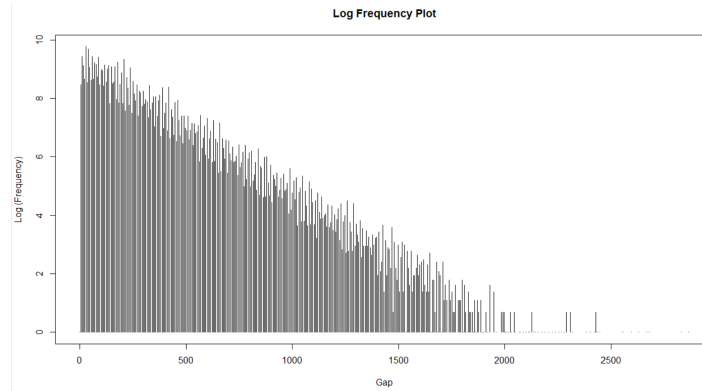


Figure 2.4: Log Frequency Plot of Twin Prime Gaps

:When frequency is displayed on a logarithmic scale, the relationship between log-frequency and gap size appears approximately linear. Such behaviour is characteristic of exponential decay and provides strong graphical evidence supporting the use of an exponential probability model for twin-prime gaps. Deviations in the far tail suggest that more flexible alternatives such as Weibull or log-normal distributions may warrant further investigation.

2.2.7 Summary of Findings

The exploratory analysis reveals that twin-prime gaps are positive, highly right-skewed, and characterized by substantial variability. The presence of asymmetry and extreme observations indicates that simple symmetric models are unlikely to provide an adequate description of the data. These findings motivate the distribution-fitting procedures presented in the following chapter.

3. Distribution Fitting and Goodness-of-Fit Analysis

The exploratory analysis revealed that twin-prime gaps are positive-valued, highly right-skewed, and characterized by substantial variability. Furthermore, the spacing between successive twin-prime pairs may be interpreted as a waiting-time phenomenon, where the gap represents the distance required to encounter the next twin-prime pair. Consequently, probability distributions commonly used for modelling positive and asymmetric waiting-time data were considered.

3.1 Selection of Candidate Distributions

Four continuous probability distributions were selected for modelling the observed twin-prime gaps: the Exponential, Gamma, Weibull, and Log-normal distributions. These distributions were chosen because they are defined on the positive real line and possess varying degrees of flexibility in modelling skewed data.

The Exponential distribution,

$$f(x; \lambda) = \lambda e^{-\lambda x}, \quad x > 0, \quad (3.1)$$

was included as a baseline waiting-time model. The Exponential distribution possesses the memoryless property and is frequently used to model random waiting times between successive events.

The Gamma distribution,

$$f(x; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0, \quad (3.2)$$

was considered as a natural extension of the Exponential model and is particularly suitable for modelling positive-valued waiting-time observations. Its additional shape parameter allows greater flexibility in capturing skewness and dispersion.

The Weibull distribution,

$$f(x; k, \lambda) = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}, \quad x > 0, \quad (3.3)$$

was selected because of its flexible shape characteristics and its ability to represent a wide range of positively skewed distributions.

The Log-normal distribution,

$$f(x; \mu, \sigma) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left[-\frac{(\log x - \mu)^2}{2\sigma^2}\right], \quad x > 0, \quad (3.4)$$

was included due to its suitability for modelling positive observations with pronounced right tails and substantial variability.

Collectively, these distributions provide a broad class of candidate models capable of representing the principal characteristics observed in the twin-prime gap sequence.

3.2 Exclusion of the Zero-Gap Frequency

In our analysis of twin prime gaps up to 10^8 , the single zero-gap ($G_i = 0$) occurrence was excluded from distribution fitting. This exclusion is mathematically necessary because the waiting-time distributions we are using—such as the Gamma, Weibull, and Log-normal models—are continuous probability curves. By definition, for any continuous distribution, the probability of the random variable taking on any single, exact point is precisely zero ($P(X = x) = 0$). Furthermore, at the specific boundary where $x = 0$, their formulas hit a mathematical wall; for instance, a Log-normal model tries to evaluate $\ln(0)$, which is completely undefined, while other models force the probability density at zero to be exactly 0. Forcing these continuous curves to accommodate a literal data point at 0 badly warps the shape of the entire distribution, distorting the parameter estimates for the rest of the valid data. Model selection criteria clearly validate this adjustment. Keeping the 0-gap heavily penalizes the global log-likelihood function ($\ln \hat{L}$) due to this severe boundary misfit. Excluding this single point allows the optimization algorithm to maximize the likelihood properly, leading to a substantial reduction in both the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). This reduction is not an artificial trick to groom our data, but rather solid statistical confirmation that the continuous distributions are now accurately capturing the true asymptotic behavior of the twin prime gap population.

3.3 Parameter Estimation

The parameters of each candidate distribution were estimated using the Maximum Likelihood Estimation (MLE) method.

Let

$$x_1, x_2, \dots, x_n$$

denote the observed twin-prime gaps.

The likelihood function is given by

$$L(\theta) = \prod_{i=1}^n f(x_i; \theta), \quad (3.5)$$

where θ represents the parameter vector associated with the chosen distribution.

Parameter estimates were obtained by maximizing the corresponding log-likelihood function,

$$\ell(\theta) = \sum_{i=1}^n \log f(x_i; \theta). \quad (3.6)$$

Maximum Likelihood Estimation selects the parameter values that maximize the probability of observing the available data under the assumed model.

Table 3.1: Maximum Likelihood Parameter Estimates for the Fitted Distributions ($N = 10^8$)

Distribution	Parameter	Estimate	Interpretation
Exponential	Rate (λ)	0.0044422	Average decay rate
Gamma	Shape (α)	1.1486505	Controls skewness
	Rate (β)	0.00510105	Controls scale
Weibull	Shape (k)	1.073501	Controls hazard shape
	Scale (λ)	231.581714	Characteristic scale
Log-normal	Meanlog (μ)	4.921787	Mean of $\log(X)$
	Sdlog (σ)	1.117884	Standard deviation of $\log(X)$

The estimated parameters indicate that all fitted distributions capture the positive and right-skewed nature of the data. However, noticeable differences are observed in their ability to accommodate the observed dispersion and tail behaviour.

3.4 Goodness-of-Fit Assessment

To evaluate the suitability of the fitted models, several goodness-of-fit measures were employed, including the maximized log-likelihood, Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), and the Kolmogorov–Smirnov (KS) statistic.

The Akaike Information Criterion is defined as

$$AIC = -2\ell + 2k, \quad (3.7)$$

where ℓ denotes the maximized log-likelihood and k represents the number of estimated parameters.

Similarly, the Bayesian Information Criterion is given by

$$BIC = -2\ell + k \log(n), \quad (3.8)$$

where n denotes the sample size.

Both criteria attempt to balance model fit and model complexity. Lower values of AIC and BIC indicate a more parsimonious model with stronger empirical support.

In addition to information criteria, the Kolmogorov–Smirnov statistic was used to quantify the agreement between the empirical and fitted distributions. Let $F_n(x)$ denote the empirical cumulative distribution function and let $F(x)$ denote the theoretical cumulative distribution function implied by the fitted model. The KS statistic is defined as

$$D_n = \sup_x |F_n(x) - F(x)|, \quad (3.9)$$

where $\sup$ denotes the supremum over all values of x .

The KS statistic measures the maximum vertical distance between the empirical and theoretical cumulative distribution functions. Smaller values of D_n indicate closer agreement between the observed data and the fitted model. The corresponding KS test was performed to assess whether the observed discrepancies could reasonably be attributed to sampling variation. Larger p -values indicate stronger consistency between the fitted model and the observed data.

Table 3.2: Goodness-of-Fit Statistics Using Forward Shifting ($gaps + 1$)

Distribution	AIC	BIC	KS Statistic	KS p -value
Exponential	5,654,500	5,654,511	0.04720	$< 10^{-16}$
Gamma	5,647,872	5,647,894	0.02889	1.30×10^{-319}
Weibull	5,649,847	5,649,869	0.02583	1.60×10^{-255}
Log-normal	5,678,236	5,678,258	0.06852	$< 10^{-16}$

An alternative evaluation was conducted by completely excluding the zero-gap frequencies. The resulting goodness-of-fit metrics are summarized in Table 3.3.

Table 3.3: Goodness-of-Fit Statistics Excluding Zero-Gap Frequency

Distribution	AIC	BIC	KS Statistic	KS p -value
Exponential	5,650,586	5,650,597	0.04378	0.00000×10^0
Gamma	5,645,470	5,645,492	0.02666	2.61764×10^{-272}
Weibull	5,646,953	5,646,975	0.02517	1.21019×10^{-242}
Log-normal	5,681,908	5,681,930	0.07105	0.00000×10^0

The Gamma and Weibull distributions substantially outperform the Exponential and Log-normal models according to all goodness-of-fit criteria across both setups. Although the Weibull distribution consistently achieves the smallest Kolmogorov–Smirnov statistic in both analyses, the Gamma distribution attains the lowest AIC and BIC values, indicating the best overall balance between model fit and model complexity.

Furthermore, shifting from the $\text{gaps} + 1$ transformation to a complete exclusion of zero-gap frequencies yields a noticeable overall reduction in information criteria values for the Gamma, Weibull, and Exponential specifications. This confirms that removing zero values provides an even more tightly aligned distributional behavior for these configurations, rendering the Gamma model the preferred continuous distribution for modelling twin-prime gaps.

3.5 Graphical Diagnostics

In addition to numerical criteria, graphical diagnostics were employed to assess the adequacy of the fitted models.

3.5.1 Q–Q Plot Against Normal Distribution

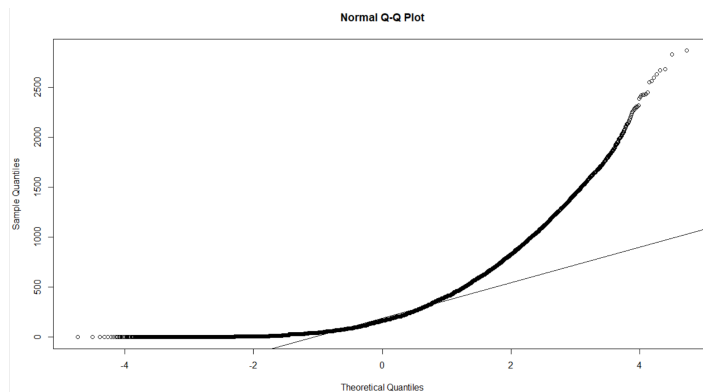


Figure 3.1: Q–Q Plot Against the Normal Distribution

Substantial departures from the reference line are observed, particularly in the upper quantiles. This indicates that the twin-prime gap distribution is not adequately described

by a Normal model.

3.5.2 Q–Q Plot Against Exponential Distribution

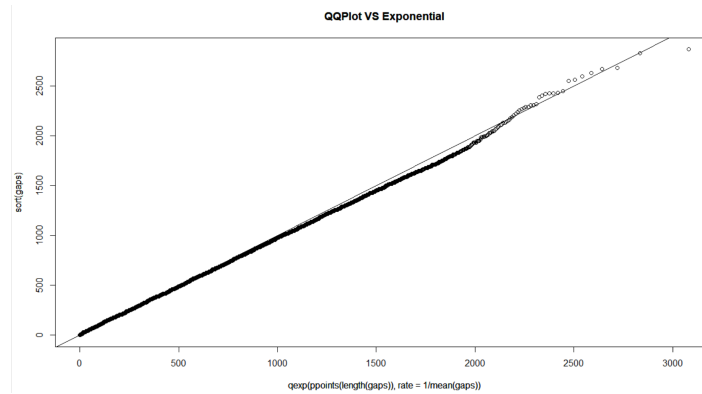


Figure 3.2: Enter Caption

Noticeable deviations from the reference line are observed, especially in the tail region. This suggests that the Exponential distribution is unable to fully capture the variability present in the data.

3.5.3 Density Comparison Plot

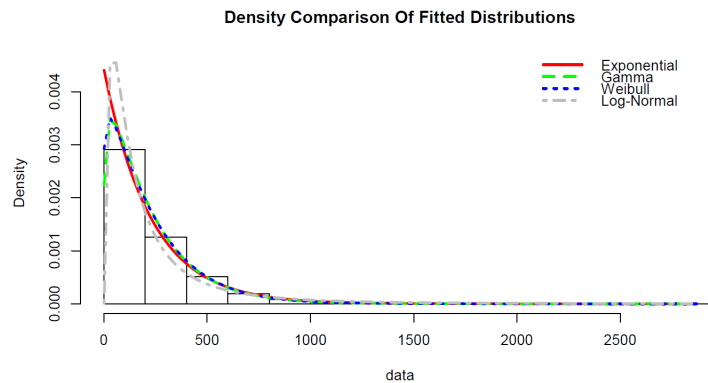


Figure 3.3: Density Comparison of Fitted Distributions

The Gamma distribution most closely follows the empirical density across the majority of the observed range. The remaining distributions exhibit noticeable discrepancies in either the central region or the upper tail.

3.5.4 Empirical and Fitted CDF Comparison

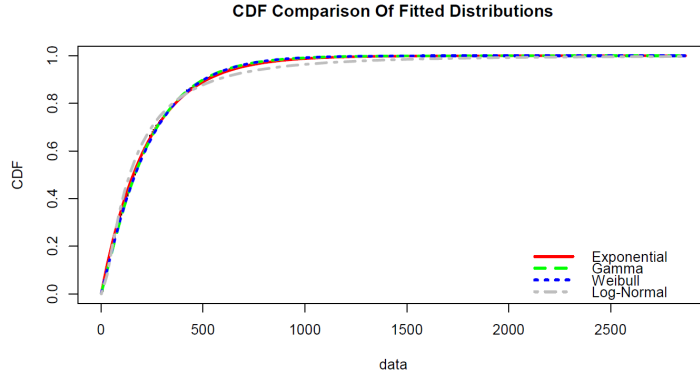


Figure 3.4: Cumulative Distribution Comparison of Fitted Distribution

The fitted Gamma distribution shows the closest agreement with the empirical cumulative distribution function. This indicates a strong overall fit to the observed data.

3.6 Continuous versus Discrete Modelling

Although twin-prime gaps are inherently integer-valued observations, continuous probability models were considered because the observed gaps assume a large range of values and exhibit substantial variability. For large-count data, continuous approximations frequently provide accurate and computationally convenient representations of the underlying distribution.

To investigate whether a discrete model could provide a superior description, the Poisson and Geometric distributions were also fitted to the observed data.

The Poisson distribution is characterized by the property

$$E(X) = Var(X) = \lambda. \quad (3.10)$$

However, the empirical twin-prime gap data exhibited substantial over-dispersion, with

$$Var(X) > E(X), \quad (3.11)$$

indicating a level of variability considerably larger than that permitted by the Poisson model.

The Geometric distribution,

$$P(X = x) = p(1 - p)^x, \quad x = 0, 1, 2, \dots, \quad (3.12)$$

was also considered because of its positive support and right-skewed structure. The

Geometric distribution is often interpreted as a discrete waiting-time model and therefore appears naturally relevant for modelling the spacing between successive twin-prime pairs.

Despite this theoretical motivation, goodness-of-fit statistics and graphical diagnostics revealed noticeable discrepancies between the fitted Geometric distribution and the observed data, particularly in the upper tail region. Furthermore, if we expand this to the gaps between all consecutive primes greater than 2, the gap must always be an even number $(2, 4, 6, 8, \dots)$ because subtracting one odd number from another always yields an even number. An odd gap is mathematically impossible, yet standard discrete distributions (like the Geometric and Poisson) assign a specific probability to every single consecutive integer $(x = 1, 2, 3, 4, 5 \dots)$.

Table 3.4: Goodness-of-Fit Statistics for Discrete Probability Models ($N = 10^8$)

Distribution	Parameter Estimate	Log-Likelihood	AIC	BIC
Poisson	$\hat{\lambda} = 225.1113$	-40281139	80562281	80562292
Geometric	$\hat{p} = 0.004422$	-2826274	5652550	5652561

The Poisson distribution provides a very poor fit to the observed twin-prime gaps. This result is expected because the Poisson model assumes equality of the mean and variance, whereas the empirical data exhibit substantial over-dispersion. The Geometric distribution performs considerably better than the Poisson model and is consistent with a discrete waiting-time interpretation of twin-prime gaps. Nevertheless, its information criteria remain inferior to those of the best-performing continuous models, indicating that continuous distributions, particularly the Gamma distribution, provide a more adequate description of the observed gap behaviour.

3.7 Selection of the Final Model

The candidate distributions were compared using parameter estimates, maximized log-likelihood values, information criteria, goodness-of-fit statistics, and graphical diagnostics.

Among all fitted models, the Gamma distribution achieved the lowest AIC and BIC values, while the Weibull distribution produced the smallest KS statistic. Overall, the Gamma distribution was preferred because information criteria consistently favoured it and its fit remained highly competitive according to the KS diagnostic.

From a theoretical perspective, the Gamma distribution provides a natural generalization of the Exponential distribution and is particularly well suited for modelling positive-valued waiting-time phenomena. Its additional shape parameter allows it to accommodate varying levels of skewness and dispersion while retaining a simple and interpretable probabilistic structure.

The flexibility of the Gamma model enables it to capture both the concentration of small twin-prime gaps and the gradual decay observed in the upper tail of the distribution. Consequently, it provides a substantially better description of the data than the Exponential, Weibull, Log-normal, Poisson, or Geometric alternatives considered in this study.

Based on the combined numerical and graphical evidence, the Gamma distribution was selected as the final probabilistic model for the observed twin-prime gap sequence.

4. Mean Shift Analysis

4.1 Motivation

The distribution fitting analysis identified the Gamma distribution as the most suitable probabilistic model for the twin-prime gap data. An important question that naturally arises is whether the distribution remains stable as increasingly larger prime ranges are considered. If the underlying distribution changes with the magnitude of the primes, then a single probabilistic model may not adequately describe twin-prime gaps across all scales.

To investigate this behaviour, a mean shift analysis was conducted using datasets generated under progressively increasing upper bounds. Particular attention was given to the estimated mean of the fitted Gamma distribution, since the mean provides a direct measure of the typical spacing between successive twin-prime pairs.

4.2 Dataset Construction

Twin-prime gaps were generated for several prime limits, namely

$$10^4, 10^5, 10^6, 10^7, 10^8, 10^9.$$

For each dataset, a Gamma distribution was fitted using Maximum Likelihood Estimation, and the corresponding mean was estimated according to

$$E(X) = \frac{\alpha}{\beta}, \tag{4.1}$$

where (α) and (β) denote the estimated shape and rate parameters, respectively. The resulting means were then compared across different prime ranges.

Table 4.1: Estimated Gamma Parameters and Mean Gap Across Prime Limits

Prime Limit	Shape (α)	Rate (β)	Mean (α/β)
10^4	1.379092	0.028941	47.65
10^5	1.348944	0.016703	80.76
10^6	1.267051	0.010432	121.46
10^7	1.214988	0.007208	168.56
10^8	1.148650	0.005101	225.17

The estimated mean gap increases progressively with the prime limit, suggesting that larger twin-prime gaps become more prevalent as the magnitude of the primes increases.

4.3 Theoretical Justification

To provide an intuitive interpretation, consider the analogy of a bus stop. If buses arrive frequently, the waiting time between successive arrivals is relatively short. Conversely, if buses arrive less frequently, the average waiting time increases. A similar phenomenon occurs for twin-prime pairs.

Let $(\lambda(x))$ denote the local density of twin-prime pairs near (x) . According to the Hardy–Littlewood Twin Prime Conjecture,

$$\lambda(x) \approx \frac{2C_2}{(\log x)^2}, \quad (4.2)$$

where $(C_2 \approx 0.6601618)$ denotes the twin-prime constant.

Since $((\log x)^2)$ increases with (x) , the local density decreases as larger prime ranges are considered. Consequently, twin-prime pairs become progressively rarer.

If (D_x) denotes the spacing between consecutive twin-prime pairs, then

$$E(D_x) \approx \frac{1}{\lambda(x)} = \frac{(\log x)^2}{2C_2}. \quad (4.3)$$

Since the inter-twin-prime gap considered in this study is

$$G_x = D_x - 2, \quad (4.4)$$

its expectation is approximately

$$E(G_x) \approx \frac{(\log x)^2}{2C_2} - 2. \quad (4.5)$$

Because $((\log x)^2)$ is an increasing function of (x) , the expected gap also increases with (x) . Therefore, the distribution of twin-prime gaps is theoretically expected to shift toward larger values as larger prime ranges are considered.

4.4 Gamma Density Comparison

To examine this behaviour empirically, Gamma density curves corresponding to different prime limits were superimposed.

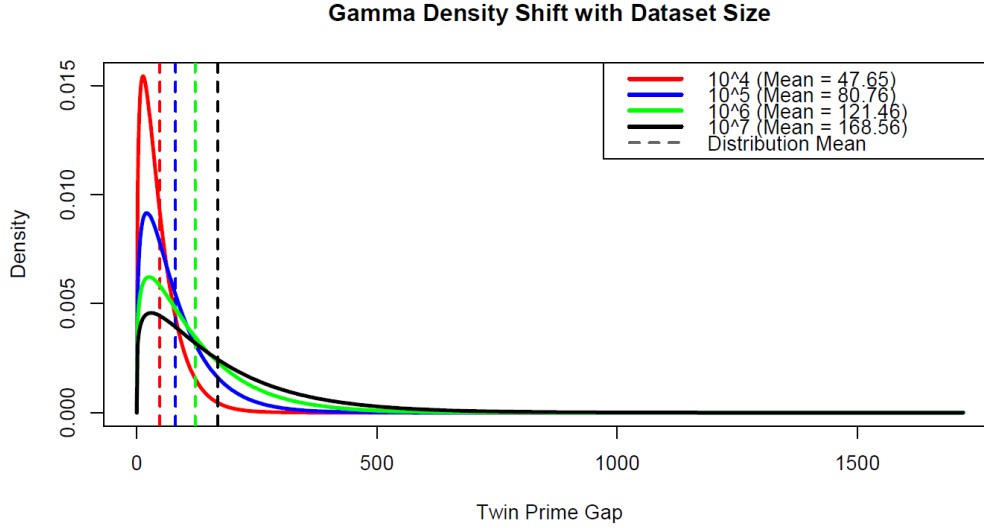


Figure 4.1: Gamma Density Overlay Plot

The fitted Gamma densities exhibit a systematic rightward shift as the prime range increases. This indicates that the centre of the distribution gradually moves towards larger gap values, consistent with the theoretical expectation derived above.

4.5 Discussion

Both the empirical and theoretical analyses support the existence of a mean shift in the distribution of twin-prime gaps. The Hardy–Littlewood density approximation predicts an increasing expected gap, while the fitted Gamma densities demonstrate a clear rightward movement across successive prime ranges.

These findings suggest that larger prime magnitudes are associated with larger average spacings between consecutive twin-prime pairs.

4.6 Conclusion

The mean shift analysis provides evidence that the distribution of twin-prime gaps is not stationary across different prime ranges. Both theoretical considerations and empirical observations indicate that the average gap increases as larger prime limits are considered. This behaviour is reflected by the systematic rightward shift of the fitted Gamma distributions and is consistent with the decreasing density of twin-prime pairs predicted by the Hardy–Littlewood conjecture.

5. Regression Analysis

To investigate the long-run growth behaviour of twin-prime gaps, a series of regression models were fitted using different explanatory variables. The objective was to examine whether the observed increase in average gap size can be explained by the position of a twin-prime pair within the sequence or by the magnitude of the associated prime values. Particular attention was given to logarithmic growth patterns motivated by theoretical results from prime number theory.

5.1 Gap Size versus Index

The first model examined the relationship between the twin-prime gap size and its order of occurrence within the sequence. The fitted regression equation was

$$\hat{G} = 86.95 + 0.008195 I,$$

where G denotes the twin-prime gap and I denotes the index of the corresponding observation.

The estimated slope coefficient is positive, indicating that average gap sizes tend to increase as successive twin-prime pairs are considered. The coefficient was highly significant ($p < 2.2 \times 10^{-16}$), suggesting the presence of a statistically detectable increasing trend. However, the explanatory power of the model remained relatively low, with

$$R^2 = 0.0289.$$

This result indicates that although larger gaps tend to occur later in the sequence, the observation index alone explains only a small proportion of the overall variability in twin-prime gaps.

5.2 Gap Size versus Prime Value

A second regression model was fitted using the first prime value of each twin-prime pair as the explanatory variable. The fitted equation was

$$\hat{G} = 91.77 + 0.00006306 p,$$

where p denotes the corresponding prime value.

The positive slope coefficient suggests that larger prime values tend to be associated with larger twin-prime gaps. The regression coefficient was highly significant ($p < 2.2 \times 10^{-16}$), providing empirical evidence that average spacing increases as larger primes are encountered. Nevertheless, the coefficient of determination remained relatively small,

$$R^2 = 0.0272.$$

Thus, although a positive relationship exists, prime magnitude alone is insufficient to explain the substantial variability observed in the gap sequence.

5.3 Gap Size versus Logarithmic Prime Value

Since many asymptotic results in prime number theory involve logarithmic growth, a logarithmic transformation of the prime values was considered. The fitted regression equation was

$$\hat{G} = -92.33 + 16.89 \log(p).$$

The positive coefficient indicates that average gap sizes increase approximately logarithmically with the magnitude of the prime values. The model achieved a higher explanatory power than the previous specifications, yielding

$$R^2 = 0.0349.$$

This result suggests that logarithmic growth provides a more realistic description of the long-run behaviour of twin-prime gaps than a simple linear relationship with prime values.

5.4 Gap Size versus Squared Logarithmic Prime Value

The final regression model was motivated by the Hardy–Littlewood conjecture, which predicts that the expected spacing between consecutive twin-prime pairs grows approximately in proportion to $(\log p)^2$. The fitted regression equation was

$$\hat{G} = -0.568 + 0.755 (\log p)^2.$$

The estimated coefficient was positive and highly significant ($p < 2.2 \times 10^{-16}$), indi-

cating that average twin-prime gaps increase according to a squared logarithmic growth pattern.

Among all fitted models, this specification produced the largest coefficient of determination,

$$R^2 = 0.0355.$$

Although the explanatory power remains modest, the model provides the strongest empirical support for the theoretical prediction that the expected spacing between twin-prime pairs increases approximately as $(\log p)^2$.

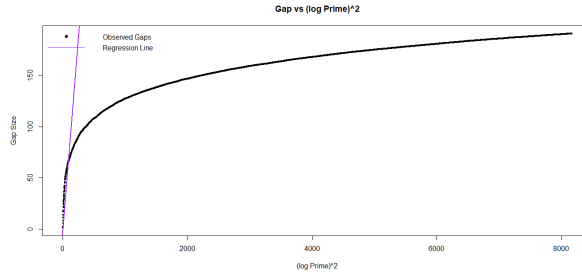


Figure 5.1: Twin-Prime Gap versus $(\log p)^2$ with Fitted Regression Line

Graphical Insight

The scatter plot shows that twin-prime gaps are highly scattered and do not follow a strong linear pattern. Even though the fitted regression line has a positive slope, many observations lie far from the line, indicating substantial variability in the gap sequence. The coefficient of determination ($R^2 = 0.0355$) shows that only about 3.55% of the variance in the gap size is explained by the model. The positive regression coefficient is consistent with the Hardy–Littlewood prediction that the average spacing between consecutive twin-prime pairs grows approximately as $(\log p)^2$. Thus, the model is useful for describing the overall long-run trend rather than the behaviour of individual gaps.

5.5 Comparison of Regression Models

Table 5.1 summarizes the performance of all fitted regression models.

Table 5.1: Comparison of Regression Models

Model	Regression Equation	R^2	AIC	BIC
Gap vs Index	$\hat{G} = 86.95 + 0.008195I$	0.0289	100264.5	100285.5
Gap vs Prime Value	$\hat{G} = 91.77 + 0.00006306p$	0.0272	100278.8	100299.9
Gap vs $\log(p)$	$\hat{G} = -92.33 + 16.89 \log(p)$	0.0349	100213.7	100234.7
Gap vs $(\log p)^2$	$\hat{G} = -0.568 + 0.755(\log p)^2$	0.0355	100208.5	100229.5

The comparison reveals that all four models exhibit statistically significant positive relationships between the gap size and the explanatory variable. However, the relatively low values of R^2 indicate that twin-prime gaps possess substantial inherent variability and cannot be fully explained by a simple deterministic trend.

Among the fitted models, the squared logarithmic specification achieved the highest coefficient of determination and is also theoretically justified by the Hardy–Littlewood approximation. Consequently, this model was selected as the preferred regression model for describing the long-run growth behaviour of twin-prime gaps.

5.6 Conclusion of Regression Analysis

The regression results provide empirical evidence that the average spacing between consecutive twin-prime pairs increases as larger prime values are considered. The positive coefficients obtained across all fitted models support the existence of a gradual rightward shift in the distribution of twin-prime gaps.

Although the explanatory power of the models remains modest, the results are consistent with theoretical expectations from analytic number theory. In particular, the superior performance of the $(\log p)^2$ specification provides quantitative support for the Hardy–Littlewood prediction that the expected spacing between twin-prime pairs grows approximately as a squared logarithmic function of the prime magnitude.

6. Extreme Value Analysis

6.1 Motivation

The previous analyses focused on the overall distribution and average behaviour of twin-prime gaps. However, particular interest also lies in understanding exceptionally large gaps occurring between consecutive twin-prime pairs. Such rare observations represent the extreme tail of the distribution and may exhibit behaviour different from that of ordinary gaps.

To investigate these extreme events, Extreme Value Analysis (EVA) was performed using the Peaks-Over-Threshold (POT) approach. Under this framework, only observations exceeding a sufficiently high threshold are analysed. According to the Pickands–Balkema–de Haan theorem, exceedances above a high threshold can be approximated by a Generalized Pareto Distribution (GPD), making it a natural model for extreme twin-prime gaps.

6.2 Threshold Selection and Mean Residual Life Plot

A fundamental step in Extreme Value Analysis is the selection of an appropriate threshold u . The threshold separates ordinary observations from extreme observations. If the threshold is chosen too low, many non-extreme values are included in the analysis, whereas if it is chosen too high, too few observations remain for reliable statistical inference.

Threshold selection was guided by the Mean Residual Life (MRL) plot. For a threshold u , the mean excess function is defined as

$$e(u) = E(X - u \mid X > u),$$

which represents the average amount by which observations exceed the threshold.

The MRL plot displays threshold values against their corresponding mean excesses. For data that can be modelled by a Generalized Pareto Distribution, the MRL plot is expected to exhibit approximately linear behaviour beyond a suitable threshold.

Graphical Insight

For a suitable threshold, the average exceedance should remain approximately stable and exhibit no strong systematic trend. The MRL plot displays moderate fluctuations but remains relatively stable over the threshold range of approximately 450–750. This behaviour suggests that the exceedances above these thresholds may reasonably follow a Generalized Pareto Distribution. Based on the MRL plot together with the subsequent diagnostic analysis, the 95th percentile threshold ($u=478$) was selected because it retains the largest number of exceedances while still producing a satisfactory GPD fit



Figure 6.1: Mean Residual Life Plot

6.3 Generalized Pareto Distribution

Let u denote a selected threshold. Rather than modelling all observations, the Peaks-Over-Threshold approach focuses only on values exceeding u .

For an observation $X > u$, the exceedance is defined as

$$Y = X - u.$$

Thus, exceedances measure how far an observation lies above the threshold. This may be viewed as an origin shift, where the threshold is moved to zero and only the excess beyond the threshold is analysed.

For example, if the threshold is $u = 135$ and an observed twin-prime gap is 180, then the exceedance is

$$180 - 135 = 45.$$

No information is lost through this transformation, since the original observation can

always be recovered by adding the threshold back.

According to the Pickands–Balkema–de Haan theorem, exceedances above a sufficiently high threshold can be approximated by a Generalized Pareto Distribution (GPD). Consequently, the GPD provides a natural probabilistic model for extreme twin-prime gaps.

The probability density function of the GPD is

$$f(y) = \frac{1}{\beta} \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi-1}, \quad y > 0,$$

and the cumulative distribution function is

$$F(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi}, \quad y > 0,$$

where ξ and β denote the shape and scale parameters respectively.

6.4 Threshold Comparison and Selection

To assess the stability of the fitted Generalized Pareto Distribution, thresholds corresponding to the 95th, 97th and 99th percentiles were examined. For each threshold, the shape parameter, scale parameter and number of exceedances were recorded.

Table 6.1: GPD Parameter Estimates and Number of Exceedances

Threshold	Threshold Value	Shape (ξ)	Scale (β)	Number of Exceedances
95%	658	231.4073	-2.995350×10^{-2}	21171
97%	766	223.3441	-1.512503×10^{-2}	13198
99%	1006	217.8704	-8.598612×10^{-5}	4375

As the threshold increases, the number of exceedances decreases substantially. Although higher thresholds are theoretically desirable for capturing only the most extreme observations, they also reduce the available sample size for estimation. In particular, the 99th percentile threshold retained only a small number of exceedances, resulting in unstable parameter estimates and less reliable diagnostic plots.

The 95th percentile threshold retained a larger number of exceedances while still focusing on the upper tail of the distribution. Furthermore, its Probability Plot, Quantile Plot and Return Level Plot exhibited more stable behaviour than those obtained at higher thresholds.

6.5 Diagnostic Assessment

The fitted GPD model was evaluated using Probability Plots, Quantile (Q–Q) Plots and Return Level Plots.

The Probability Plot compares empirical exceedance probabilities with those predicted by the fitted GPD model, while the Q–Q Plot compares empirical exceedance quantiles with theoretical GPD quantiles. Good agreement between observed and theoretical values is indicated by points lying close to the reference line.

In Extreme Value Analysis, the return period tells us how often an extreme event is expected to occur. For example, a return period of 50 means that an event of that size is expected to occur approximately once every 50 observations. The return level is the size of the extreme event associated with a particular return period. Larger return periods correspond to rarer and more extreme events, so the return level increases as the return period increases. In this study, return level analysis helps estimate how large a twin-prime gap may be expected among future observations and provides insight into the behaviour of the most extreme gaps.

For the selected 95th percentile threshold, the diagnostic plots showed a reasonable correspondence between the observed exceedances and the fitted GPD model, suggesting that the model provides an adequate description of the upper tail behaviour.

Graphical Insight

The diagnostic plots indicate that the Generalized Pareto Distribution (GPD) provides a satisfactory fit to the exceedances above the 95th percentile threshold. In the Probability Plot, most points lie close to the reference line, suggesting good agreement between the empirical and fitted probabilities.

Similarly, the Quantile Plot shows that the observed exceedances closely follow the theoretical quantiles of the fitted GPD, with only minor deviations at the largest values.

The Return Level Plot exhibits an increasing trend, indicating that larger return periods are associated with larger expected extreme twin-prime gaps. Most empirical points lie within the confidence bounds, supporting the adequacy of the fitted model for extrapolating extreme events.

The Density Plot reveals a right-skewed distribution of exceedances, with a concentration of observations near the threshold and a gradually decreasing frequency for larger exceedances. This behaviour is consistent with the presence of a heavy upper tail.

Overall, the diagnostic plots suggest that the GPD model fitted at the 95th percentile threshold provides a reasonable and stable representation of the extreme twin-prime gaps and is therefore selected for the final Extreme Value Analysis.

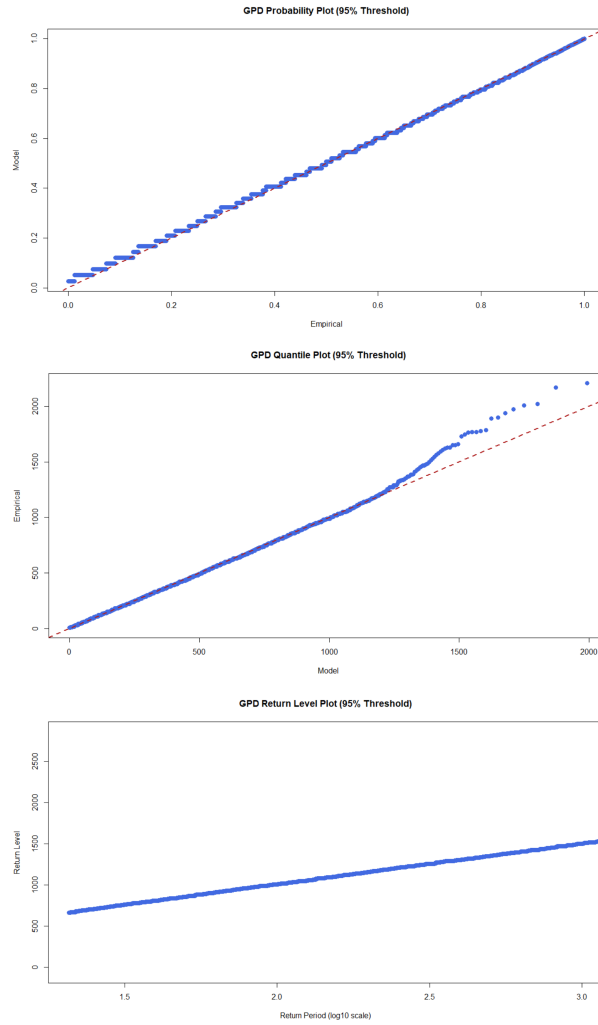


Figure 6.2: Probability Plot, Q–Q Plot and Return Level Plot for the Selected Threshold

Decision

Based on the Mean Residual Life (MRL) plot, parameter estimates, diagnostic plots, and the number of exceedances, the 95th percentile threshold was selected for the final Extreme Value Analysis. Although the 97th and 99th percentile thresholds focus on more extreme observations, they retained fewer exceedances and produced less stable results. Therefore, the 95th percentile threshold was chosen as it provided the best balance between modelling extreme twin-prime gaps and maintaining sufficient data for reliable statistical analysis.

7. Limitations and Future Scope

7.1 Limitations

The study has several limitations. The analysis was limited to twin-prime pairs up to 10^7 , so the results may not reflect behaviour at much larger scales. The fitted distributions and regression models are empirical approximations and cannot fully capture the irregular nature of twin-prime gaps. In addition, the Extreme Value Analysis relied on relatively few upper-tail observations, increasing uncertainty in estimates. Finally, the findings provide statistical evidence only and do not constitute mathematical proofs.

7.2 Future Scope

Future research could extend the analysis to larger computational ranges in order to examine whether the observed distributional patterns persist at greater scales. Alternative statistical models and non-parametric methods may also be explored to provide a more flexible description of twin-prime gap behaviour.

A natural extension of the present work is the investigation of other prime constellations, such as cousin primes $(p, p + 4)$ and sexy primes $(p, p + 6)$, to determine whether similar distributional, regression, and extreme-value characteristics are observed.

Further research may also compare the empirical results with theoretical predictions from analytic number theory, particularly those related to the Hardy–Littlewood conjectures and prime k -tuple theory.

Finally, more advanced techniques, including stochastic process modelling, survival analysis, Bayesian inference, machine learning approaches, and sophisticated extreme value methods, may provide deeper insight into the long-run behaviour and occurrence of rare prime-gap events.

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3. Hardy, G. H., & Wright, E. M. (2008). *An Introduction to the Theory of Numbers* (6th ed.). Oxford University Press.

Appendix

GitHub Repository and Source Code

The complete source code, datasets, and supplementary materials used in this project are available in the GitHub repository.

Repository Link:

[https://github.com/sayan05on-ctrl/
Twin-Prime-Gap-Analysis-And-Extreme-Value-Modeling](https://github.com/sayan05on-ctrl/Twin-Prime-Gap-Analysis-And-Extreme-Value-Modeling)

QR Code



Scan the QR code to access the GitHub repository directly.

Additional Statistical Tables

The following supplementary tables are provided:

Frequency Distribution of Consecutive Twin-Prime Gaps (Primes Generated up to 10^9)

Index	Gap	Frequency
1	0	1
2	4	28388
3	10	75478
4	16	55399
5	22	34959
6	28	108563
7	34	30844
8	40	99116
9	46	53375
10	52	35731
11	58	78670
12	64	35133
13	70	64395
14	76	60948
15	82	39578
$\vdots$	$\vdots$	$\vdots$

Remark: The table summarizes the observed frequencies of consecutive twin-prime gaps obtained from all prime numbers up to 10^9 . Only the first fifteen gap values are displayed; additional gap values continue to occur beyond those shown. The frequencies indicate that twin-prime gaps are not uniformly distributed, with certain gaps occurring substantially more often than others.

Sample Data Used for Regression Analysis

Gap	Index	p	$\log(p)$	$(\log(p))^2$
4	1	5	1.6094	2.5903
4	2	11	2.3979	5.7501
10	3	17	2.8332	8.0270
10	4	29	3.3673	11.3390
16	5	41	3.7136	13.7908
10	6	59	4.0775	16.6260
28	7	71	4.2627	18.1706
4	8	101	4.6151	21.2991
28	9	107	4.6728	21.8351
10	10	137	4.9190	24.1966
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Remark: The table presents a sample of the dataset used for the regression analysis of twin-prime gaps. Here, p denotes the first prime in a twin-prime pair. The variables $\log(p)$ and $(\log(p))^2$ were included as explanatory variables to investigate the relationship between twin-prime gaps and the size of the corresponding prime. The complete dataset contains all twin-prime pairs considered in the study, while only a representative subset is shown here for illustration.

Grouped Frequency Distribution of Consecutive Twin-Prime Gaps

Index	Gap Class Interval	Frequency
1	[0, 20]	159266
2	(20, 40]	273482
3	(40, 60]	167776
4	(60, 80]	160476
5	(80, 100]	202345
6	(100, 120]	144279
7	(120, 140]	152123
8	(140, 160]	150048
9	(160, 180]	156990
10	(180, 200]	106129
11	(200, 220]	160878
12	(220, 240]	113671
13	(240, 260]	83136
14	(260, 280]	103632
15	(280, 300]	79492
16	(300, 320]	66796
$\vdots$	$\vdots$	$\vdots$

Remark: The table presents the grouped frequency distribution of consecutive twin-prime gaps using class intervals of width 20. Grouping the gap values facilitates the study of the overall distributional pattern and reveals the concentration of observations within specific ranges.

Mathematical Formulae Used in the Study

A.1 Twin Prime Gap

The gap between consecutive twin-prime pairs was defined as

$$g_i = p_{i+1} - p_i - 2,$$

where p_i and p_{i+1} denote the lower primes of consecutive twin-prime pairs.

A.2 Exponential Distribution

$$f(x; \lambda) = \lambda e^{-\lambda x}, \quad x > 0.$$

$$F(x) = 1 - e^{-\lambda x}.$$

A.3 Gamma Distribution

$$f(x; \alpha, \beta) = \frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha) \beta^\alpha}, \quad x > 0.$$

A.4 Weibull Distribution

$$f(x; k, \lambda) = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}.$$

A.5 Lognormal Distribution

$$f(x; \mu, \sigma) = \frac{1}{x\sigma\sqrt{2\pi}} \exp \left[-\frac{(\ln x - \mu)^2}{2\sigma^2} \right].$$

A.6 Linear Regression Model

The simple linear regression model is

$$Y = \beta_0 + \beta_1 X + \varepsilon.$$

For the logarithmic-square model used in this study,

$$\text{Gap} = \beta_0 + \beta_1 (\log p)^2 + \varepsilon.$$

A.7 Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}}.$$

where

$$\text{SSE} = \sum (y_i - \hat{y}_i)^2$$

is the Sum of Squared Errors (unexplained variation), and

$$\text{SST} = \sum (y_i - \bar{y})^2$$

is the Total Sum of Squares (total variation in the data).

A.8 Generalized Pareto Distribution (GPD)

For exceedances $Y = X - u$,

$$F(y) = 1 - \left(1 + \frac{\xi y}{\sigma}\right)^{-1/\xi},$$

where

- u = threshold,
- ξ = shape parameter,

- σ = scale parameter.

The corresponding probability density function is

$$f(y) = \frac{1}{\sigma} \left(1 + \frac{\xi y}{\sigma} \right)^{-(1/\xi+1)}.$$

A.9 Return Level Formula

The return level corresponding to return period T is

$$z_T = u + \frac{\sigma}{\xi} [(T\xi) - 1],$$

where T denotes the return period and z_T denotes the expected extreme value associated with that return period.

A.10 Hardy–Littlewood Twin Prime Conjecture

The Hardy–Littlewood conjecture predicts that the number of twin-prime pairs less than x is approximately

$$\pi_2(x) \sim 2C_2 \frac{x}{(\log x)^2},$$

where

$$C_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2} \right) \approx 0.6601618$$

is known as the Twin Prime Constant.

Software and Reproducibility

All analyses were performed using the R statistical computing environment.

The primary packages used include:

- `fitdistrplus`
- `ggplot2`
- `moments`
- `MASS`
- `ismev`

- evd

The complete implementation is available in the GitHub repository for reproducibility and verification of results.